SIMATS ENGINEERING ASSESSMENT

TOOL 3

V. Sandeep Reddy

192325096

Marks:90

COURSE NAME: INTERNET PROGRAMMING **COURSE CODE: CSA43**

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

```
<!DOCTYPE html>

<html lang="en">

<head>
```

```
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Modern Web Application</title>
    <style>        body {            font-
family: Arial, sans-serif;
margin: 0;            padding: 0;
background-color: #f4f4f4;
        }
        /* Semantic Navigation Styling */
        nav {            background-
color: #333;            color:
white;            padding: 1rem;
text-align: center;
        }        nav a {
color: white;
margin: 0 15px;
text-decoration:
none;            font-
weight: bold;
        }
```

```
.container { max-width: 800px;
margin: 20px auto; background:
white; padding: 20px; border-
radius: 8px; box-shadow: 0 0 10px
rgba(0,0,0,0.1);
}
/* Video Styling */
.promo-video { text-
align: center; margin-
bottom: 30px;
}
video { max-
width: 100%;
border-radius: 5px;
}
/* Secure Form Styling */
form { display:
flex; flex-direction:
column; gap: 15px;
} input {
padding: 10px;
```

```
border: 1px solid #ccc;

border-radius: 4px;

      }

      button {           padding: 10px;

background-color: #28a745;

color: white;           border: none;

border-radius: 4px;           cursor:

pointer;

      }

   </style>

</head>

<body>

   <!-- Q3: Semantic Navigation Element -->

   <nav>

      <a href="#home">Home</a>

      <a href="#about">About</a>

      <a href="#register">Register</a>

   </nav>

   <div class="container">

      <!-- Q1: HTML tag for video embedding -->
```

```
<section class="promo-video" id="home">
  <h2>Welcome to Our Startup</h2>
  <video src="video.mp4" controls width="600">
    Your browser does not support the video tag.
  </video>
</section>

<!-- Q2: Form Submission using POST -->
<section id="register">
  <h2>Secure Registration Form</h2>
  <form action="/submit-registration" method="POST">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required>  <label for="email">Email:</label>
    <input type="email" id="email" name="email" required>

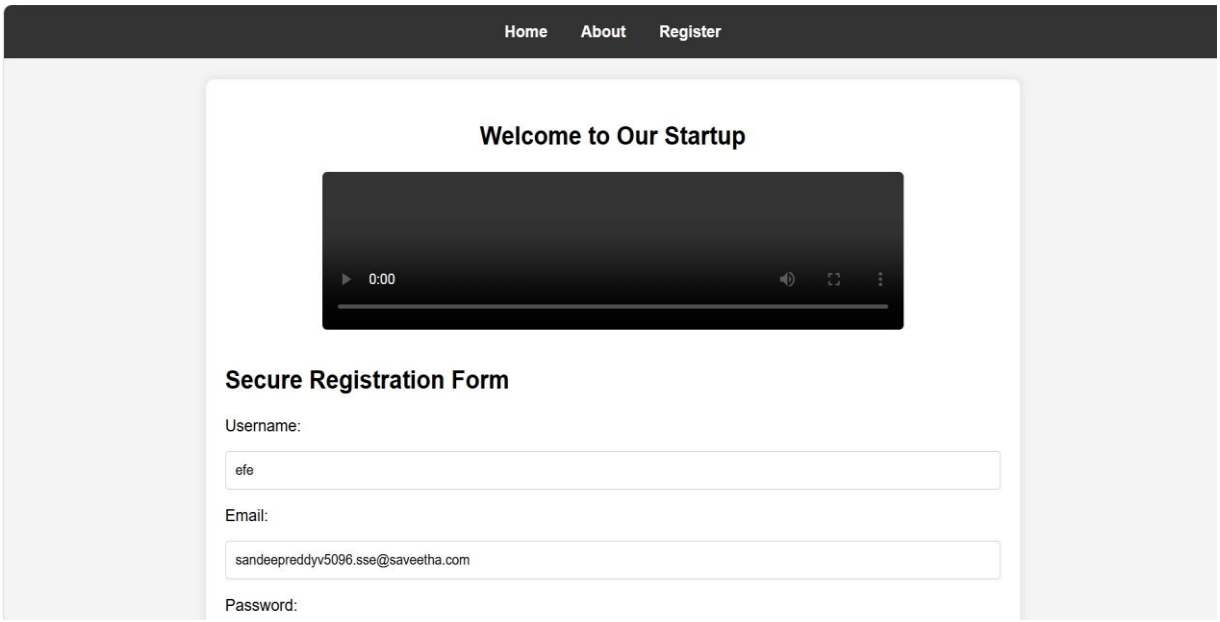
    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required>

    <button type="submit">Register securely</button>
  </form>
</section>
```

```
</div>


</body>

</html>
```



Code of Conduct:

I _V. Sandeep Reddy(192325096)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *V. Sandeep Reddy*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understanding of Case	25 22.5	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25 22.5	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20 18	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20 18	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10 9	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand